

EARLY STROKE PREDICTION USING MACHINE LEARNING

A Machine Learning Approach and website for Early Detection of Stroke Risk

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INTRODUCTION

- Stroke is one of the leading causes of death and long-term disability worldwide.
- It occurs when the blood supply to the brain is interrupted or reduced.
- Early detection of stroke risk can significantly improve survival and recovery outcomes.
- Healthcare datasets contain valuable information about patient health conditions and lifestyle habits.
- Machine learning models can assist in identifying relationships between multiple risk factors.
- This project investigates the use of machine learning for predicting early stroke risk using healthcare data.
 - A web-based deployment of the prediction model enables real-time clinical use

PROBLEM STATEMENT

- Stroke often occurs suddenly without clear warning signs.
- Healthcare datasets contain multiple factors that influence stroke risk.
- Analyzing these factors manually can be difficult due to the complexity of medical data.
- Machine learning algorithms can identify patterns that may not be visible through traditional analysis.
- An intelligent system is required to analyze patient data and predict stroke probability.
- Therefore, this project aims to develop a machine learning model for stroke prediction and develop a website.

OBJECTIVES

- To identify the early stroke risk detection system.
- To analyze the influence of clinical features using explainable AI (feature importance)
- To implement feature engineering techniques to improve model performance.
- To handle class imbalance using SMOTE.
- To compare the performance of different models and identify the most effective approach
- To deploy the best-performing model as a web application for real-time stroke risk prediction

LITERATURE REVIEW

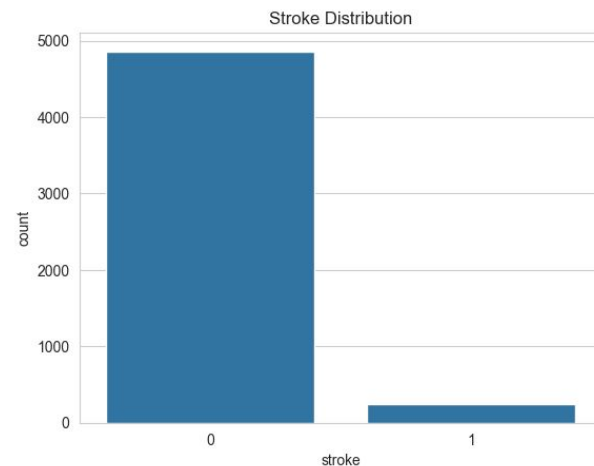
Authors	Year	Key Contribution
Dev et al.	2022	Used PCA and NN for stroke prediction; identified age, glucose, heart disease as top features.
Hassan et al.	2024	Proposed Dense Stacking Ensemble (DSE) with SMOTE and MICE imputation.
Chakraborty et al.	2024	PCA + stacking ensemble (RF+DT+KNN); 98.6% accuracy on balanced data
Melnykova et al.	2025	Demonstrated accuracy paradox on imbalanced data; RF with grid search; honest evaluation

Research Gap:

- Existing studies lack real-world deployment of stroke prediction models into usable clinical applications.
- Absence of Advanced Feature Engineering

DATASET DESCRIPTION

- Publicly available clinical dataset from Kaggle (McKinsey Healthcare Hackathon)
- The dataset contains records of 5,110 patients with health and demographic information
- Dataset includes 12 features that influence stroke occurrence
- Target variable is binary: stroke (1) or no stroke (0)
- Dataset is highly imbalanced — 4,861 non-stroke cases vs 249 stroke cases (approximately 19:1 ratio)
- Input features include age, BMI, average glucose level, hypertension, heart disease, lifestyle factors like marital status, residential type, work type and smoking status.
- Data preprocessing and feature engineering were performed before model train.



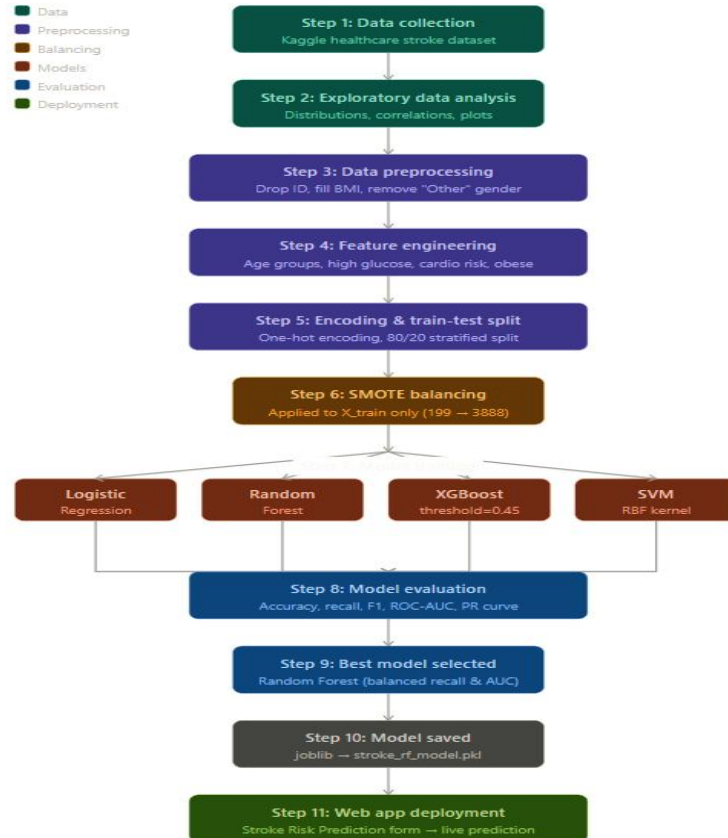
INPUT FEATURES USED

- **Patient demographics:** Age, Gender, BMI
- **Medical history:** Hypertension (yes/no), Heart Disease (yes/no)
- **Lifestyle information:** Smoking Status, Ever Married, Work Type, Residence Type
- **Clinical measurements:** Average Glucose Level
- **Engineered features:** Age Category (Young/Adult/Senior/High Risk), High Glucose flag, Cardio Risk flag, Obese flag
- **Target variable:** Stroke occurrence (0 = No Stroke, 1 = Stroke)

PROPOSED METHODOLOGY

- Data collection from the publicly available Kaggle stroke dataset
- Exploratory data analysis including distribution plots, correlation heatmap, and feature-stroke relationship analysis
- Data preprocessing: drop ID column, fill missing BMI with median, remove "Other" gender category
- Feature engineering: create age categories, high glucose flag, cardio risk composite, obese indicator
- One-hot encoding of categorical variables and train-test split (80/20 stratified)
- SMOTE applied only to training data to prevent data leakage (199 → 3,888 stroke cases)
- Training four ML models: Logistic Regression, Random Forest, XGBoost, SVM
- Comparing model performance using multiple evaluation metrics
- Selection of best-performing model (Random Forest) and saving via joblib
- Deployment as a web application for real-time stroke risk prediction

SYSTEM FLOWCHART

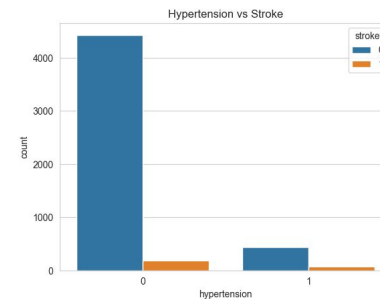
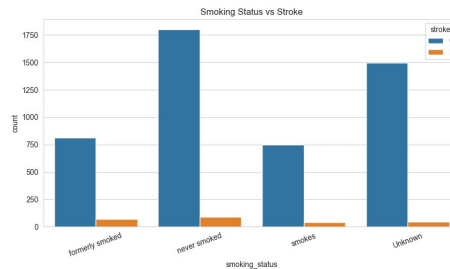
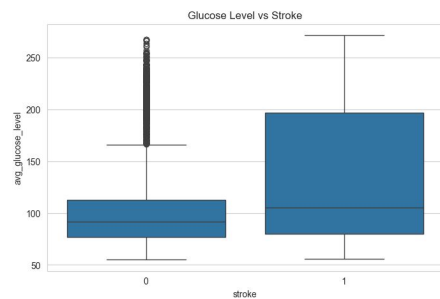
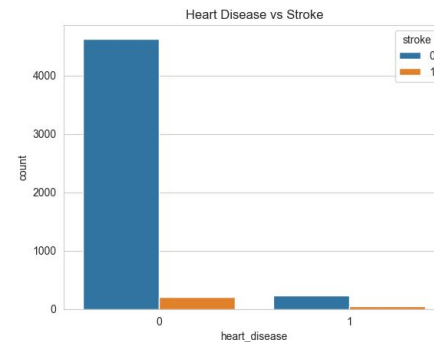
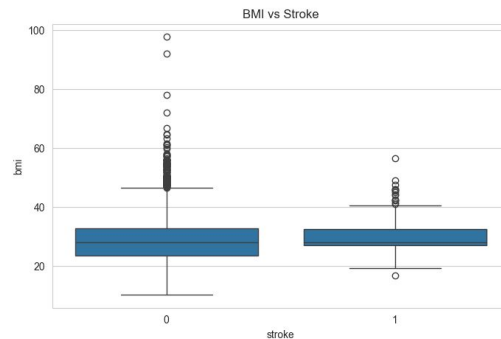
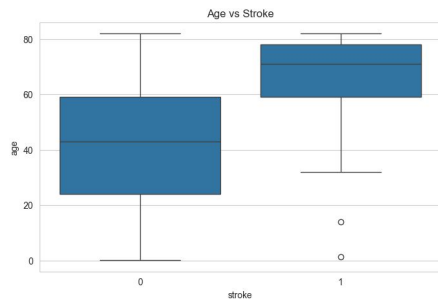


Flowchart represents the complete ML pipeline from data collection to web application deployment

EXPLORATORY DATA ANALYSIS

- Exploratory data analysis was performed to understand the distribution of features.
- Age is one of the strongest predictors of stroke risk.
- Hypertension, heart disease, smoking status were included as key predictors.
- Both glucose level and BMI are important predictive variables.
- work type and residence type did come as key predictors
- Gender and Marital status provided small correlation as the dataset was small

EXPLORATORY DATA ANALYSIS



ML MODELS USED

- **Logistic Regression:** Used as a baseline model for binary classification of stroke occurrence.
- **Random Forest:** Ensemble model to improve prediction accuracy and reduce overfitting
- **XGBoost:** Optimized boosting algorithm known for strong performance on structured datasets
- **Support Vector Machine (SVM):** Used for effective classification by finding optimal decision boundaries

MODEL EVALUATION METRICS

Accuracy: Overall correctness of prediction across all classes

Precision: Correctness of positive (stroke) predictions — avoids unnecessary false alarms

Recall: Ability to identify actual stroke cases — most critical metric in medical prediction

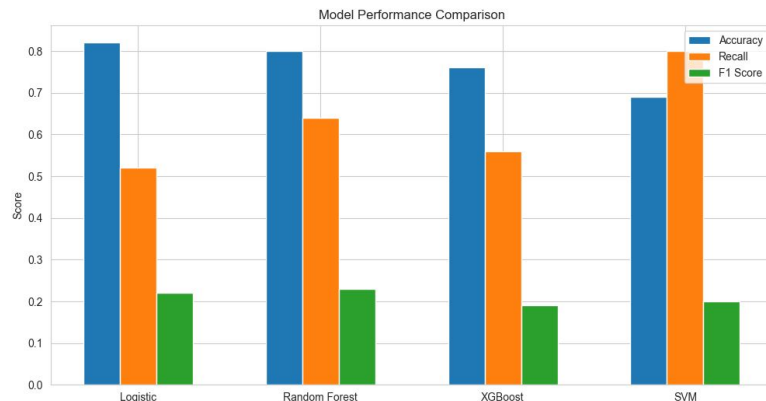
F1-Score: Harmonic mean of precision and recall — balanced assessment for imbalanced data

ROC-AUC: Ability of the model to distinguish between stroke and non-stroke cases across all thresholds

Precision-Recall curve (AP score): Especially useful for highly imbalanced datasets

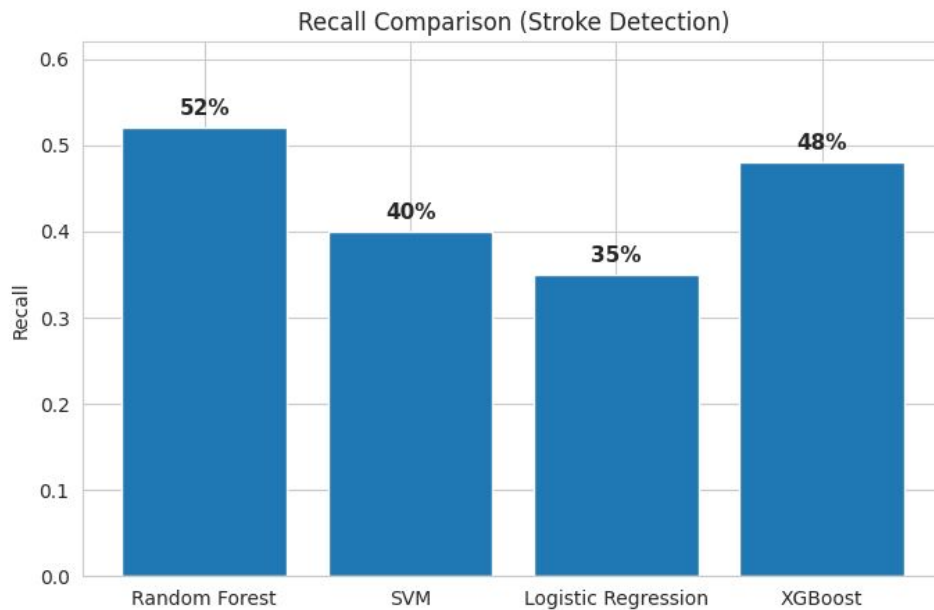
MODEL PERFORMANCE COMPARISON

- Multiple models were compared based on evaluation metrics.
- Higher Recall is needed for medical prediction.
- Random Forest achieved balanced accuracy and recall.
- SVM achieved high recall but lower precision.
- Logistic Regression provided moderate performance.
- XGBoost performed well but slightly lower than Random Forest.
- Random Forest was chosen as the final model.



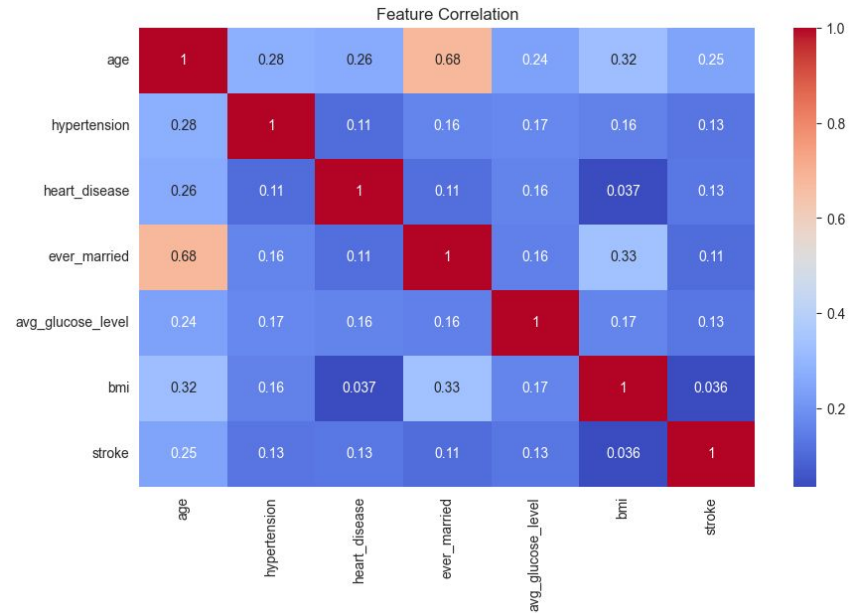
	Model	Accuracy	Recall	Precision	F1 Score
0	Logistic	0.82	0.52	0.14	0.22
1	Random Forest	0.80	0.64	0.14	0.23
2	XGBoost	0.76	0.56	0.11	0.19
3	SVM	0.69	0.80	0.11	0.20

RECALL COMPARISON



FEATURE CORRELATION ANALYSIS

- Feature correlation analysis was conducted to interpret which patient attributes most influence stroke prediction
- Correlation heatmap revealed age has the strongest correlation with stroke (0.25)
- From model analysis, the most influential features were: Age, Average Glucose Level, BMI, Hypertension, Heart Disease
- These findings align with established clinical risk factors for stroke
- This analysis improves transparency and understanding of the prediction model



RESULTS & OBSERVATION

- Multiple machine learning models were trained to predict stroke risk using patient health records
- Performance of models was compared using accuracy, recall, F1-score, ROC-AUC, and precision-recall curves
- Ensemble models such as Random Forest and XGBoost showed strong predictive performance
- Random Forest with higher recall achieved the best balance between stroke detection.
- Feature analysis confirmed that Age show significant impact on prediction results.
- SMOTE(*Synthetic Minority Over-sampling Technique*) applied correctly only to training data prevented data leakage, ensuring honest evaluation
- The study demonstrates the potential of machine learning in real-world stroke risk prediction

WEBSITE APPLICATION

- The trained Random Forest model was saved using joblib and deployed as a web application using the Flask Python framework
- Users can input patient details including gender, age, hypertension, heart disease, marital status, work type, residence type, glucose level, BMI, and smoking status
- HTML , CSS were used for structure and style of the website.
- The application provides instant real-time stroke risk prediction upon clicking "Predict".
- The web interface is accessible to healthcare practitioners for early stroke risk screening

WEBSITE APPLICATION

Stroke Risk Prediction

Enter the required details below

Gender	Age
Male	25
Hypertension	Heart Disease
No	No
Ever Married	Work Type
No	Private
Residence Type	Average Glucose Level
Urban	85
BMI	Smoking Status
24	Never smoked

PredictClear Form

Predicted Result

No Stroke Risk Flag

Prediction Verdict: No Stroke Risk Flag

Stroke Risk Prediction

Enter the required details below

Gender	Age
Female	85
Hypertension	Heart Disease
Yes	Yes
Ever Married	Work Type
Yes	Private
Residence Type	Average Glucose Level
Urban	140
BMI	Smoking Status
27	Smokes

PredictClear Form

Predicted Result

Stroke Risk Flag Detected

Prediction Verdict: Stroke Risk Flag Detected

FUTURE WORK

- Extend the study using larger and more diverse healthcare datasets from multiple institutions
- Validate the model with real clinical data in collaboration with healthcare institutions
- Incorporate diverse datasets including genetic, lifestyle, and high-tech imaging data to strengthen the model's predictive capabilities
- Explore deep learning techniques tailored for clinical interpretability and further advancements in ensemble learning methodologies
- Develop stroke subtype differentiation (ischemic vs hemorrhagic stroke) as separate classification tasks.

CONCLUSION

- Machine learning models were applied to analyze patient health records related to stroke risk prediction
- Multiple algorithms were implemented and compared to identify the best-performing model
- Random Forest emerged as the optimal model with the best balance of recall and accuracy for stroke detection
- SMOTE was correctly applied only to training data, addressing the data leakage issue present in prior studies
- The trained model was successfully deployed as a web application enabling real-time stroke risk prediction
- The web deployment contribution directly addresses a critical research gap identified across all reviewed literature

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- ❑ Hassan, A., Gulzar Ahmad, S., Ullah Munir, E. *et al.* Predictive modelling and identification of key risk factors for stroke using machine learning. *Sci Rep* 14, 11498 (2024).
<https://doi.org/10.1038/s41598-024-61665-4>

THANK YOU